

Supplementary Online Content

Cooper DJ, Nichol A, Bailey M, et al. Effect of early sustained prophylactic hypothermia on neurologic outcomes among patients with severe traumatic brain injury: the POLAR randomized clinical trial. *JAMA*. doi:10.1001/jama.2018.17075

Supplementary Results

eTables 1-7 and eFigures 1-2

Per-Protocol Analysis

eFigures 3-5 and eTables 8-9

As-Treated Analysis

eFigures 6-7 and eTables 10-11

Post Hoc Analysis

eTables 12-13

eReferences.

This supplementary material has been provided by the authors to give readers additional information about their work.

SUPPLEMENT 2 – POLAR TRIAL eTABLES, eFIGURES and eREFERENCES

This supplement has been provided by the authors to give readers additional information about their work. Supplement to: Cooper DJ, Nichol A, Bailey M, et al. Effect of Early Sustained Prophylactic Hypothermia on Neurologic Outcomes Among Patients With Severe Traumatic Brain Injury. JAMA. Published October 24, 2018. 10.1001/jama.2018.17075

1.	SUPPLEMENTARY RESULTS.....	2
	eTable 1. POLAR Trial Inclusion and Exclusion Criteria	2
	eFigure 1. Hourly Temperature and ICP During Rewarming (36 hours) in the Hypothermia Group	3
	eFigure 2. Kaplan-Meier Survival Curve	4
	eTable 2. Randomization Location and Times.....	5
	eTable 3. Protocol Violations.....	6
	eTable 4. Processes of Care	7
	eTable 5. Other Outcomes	8
	eTable 6. Adverse Events.....	9
	eTable 7. Time to Reach 33°C for Hypothermia Patients (Approximate Tertiles).....	10
2.	PER-PROTOCOL ANALYSIS	11
	eFigure 3. Patients Included in Per-Protocol Analysis.....	11
	eTable 8. Per-Protocol Baseline.....	12
	eTable 9. Per-Protocol Analysis	13
	eFigure 4. Hourly Temperature for First 4 days (96 hours) Post-Randomization in the Per-Protocol Analysis of Patients.....	14
	PER-PROTOCOL SENSITIVITY ANALYSIS	15
	eFigure 5. Patients Included in Per-Protocol – Sensitivity Analysis.....	15
	PER-PROTOCOL SENSITIVITY ANALYSIS – PRIMARY OUTCOME	15
3.	AS-TREATED ANALYSIS.....	16
	eFigure 6. Patients Included in As-Treated Analysis.....	16
	eTable 10. As-Treated Baseline	17
	eTable 11. As-Treated Analysis.....	18
	AS-TREATED SENSITIVITY ANALYSIS	19
	eFigure 7. Patients Included in As-Treated – Sensitivity Analysis	19
	AS-TREATED SENSITIVITY ANALYSIS – PRIMARY OUTCOME	19
4.	POST-HOC ANALYSES.....	20
	POST-HOC SENSITIVITY ANALYSIS OF MISSINGNESS.....	20
	eTable 12. Sensitivity Analysis of Missingness	20
	POST-HOC ANALYSES OF EVALUABLE HYPOTHERMIA PATIENTS THAT RECEIVED ADEQUATE COOLING VS NORMOTHERMIA	21
	eTable 13. Comparison of Baseline Characteristics for Patients who Completed the Hypothermia Protocol and had Primary Outcome vs All Controls with Primary Outcome	22
5.	eREFERENCES.....	23

For further information, please refer to *Supplements 1 (Protocol, Statistical Analysis Plan, and Data Safety Monitoring Committee Stopping Rules and Reports)* and *3 (Data Sharing Statement)*.

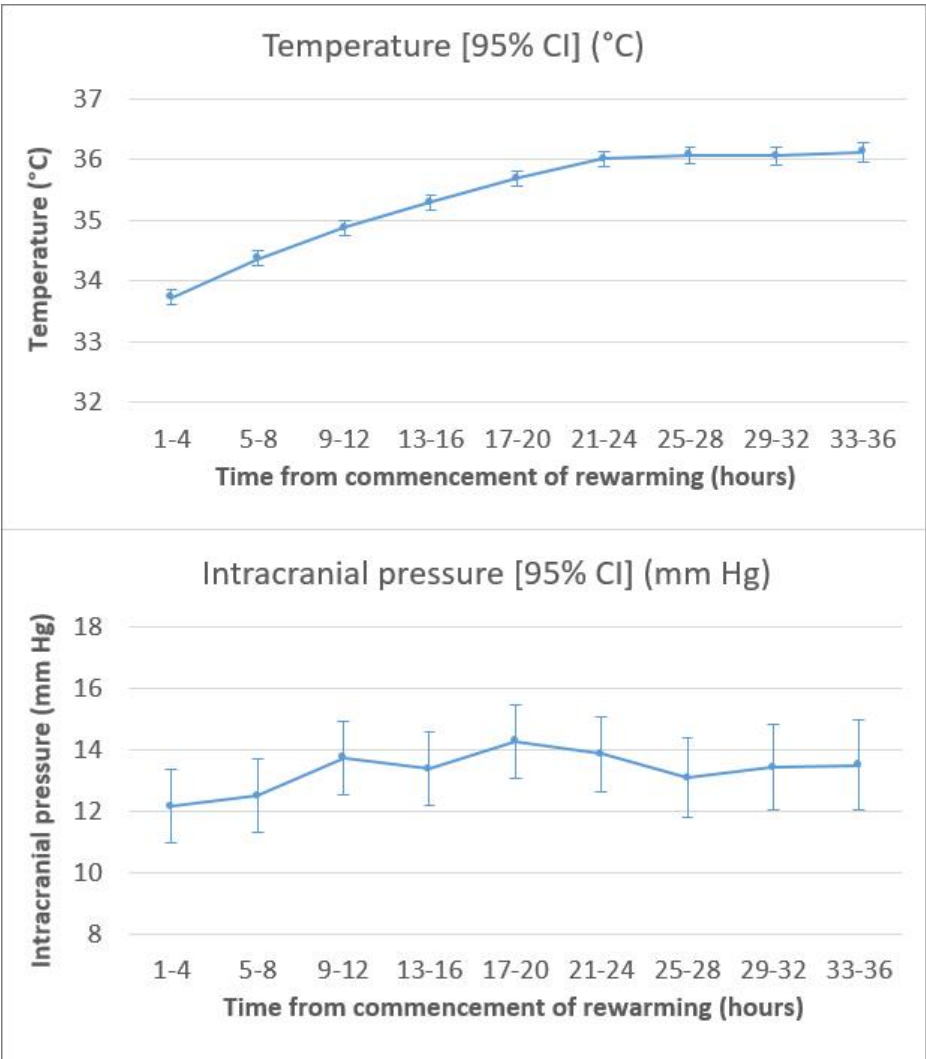
1. SUPPLEMENTARY RESULTS

eTable 1. POLAR Trial Inclusion and Exclusion Criteria

Out-of-hospital and Emergency Department inclusion criteria:	• Blunt trauma with clinical diagnosis of severe TBI and GCS score <9 • Estimated age ≥18 and <60 years of age • The patient is intubated or intubation is imminent
Out-of-hospital exclusion criteria:	• Clinical diagnosis of drug or alcohol intoxication as predominant cause of coma • Randomization unable to be performed within 3 hours of estimated time of injury • Estimated transport time to study hospital >2.5 hours • Able to be intubated without drugs • Systolic blood pressure <90 mm Hg • Heart rate >120 bpm • Cardiac arrest • GCS score = 3 + unreactive pupils • Penetrating neck/torso injury • Known or obvious pregnancy • Receiving hospital is not a study site • Evidence of current anti-coagulant treatment • Known to be carer dependent due to a pre-existing neurological condition
Emergency Department exclusion criteria:	• Clinical diagnosis of drug or alcohol intoxication as predominant cause of coma • Randomization unable to be performed within 3 hours of estimated time of injury • Able to be intubated without drugs • Persistent Systolic blood pressure <90 mm Hg • GCS score = 3 + unreactive pupils • Cardiac arrest at the scene or in transit • Clinically significant bleeding likely to require hemostatic intervention, for example: ○ Bleeding into the chest, abdomen or retro-peritoneum likely to require surgery +/- embolization ○ Pelvic fracture likely to require surgery +/- embolization ○ More than two long bone fractures requiring operative fixation • Penetrating neck/torso injury • Positive urine or blood pregnancy test • Evidence of current anti-coagulant treatment • Known to be carer dependent due to a pre-existing neurological condition • In the treating clinician's opinion, "cooling" is not in the patient's best interest

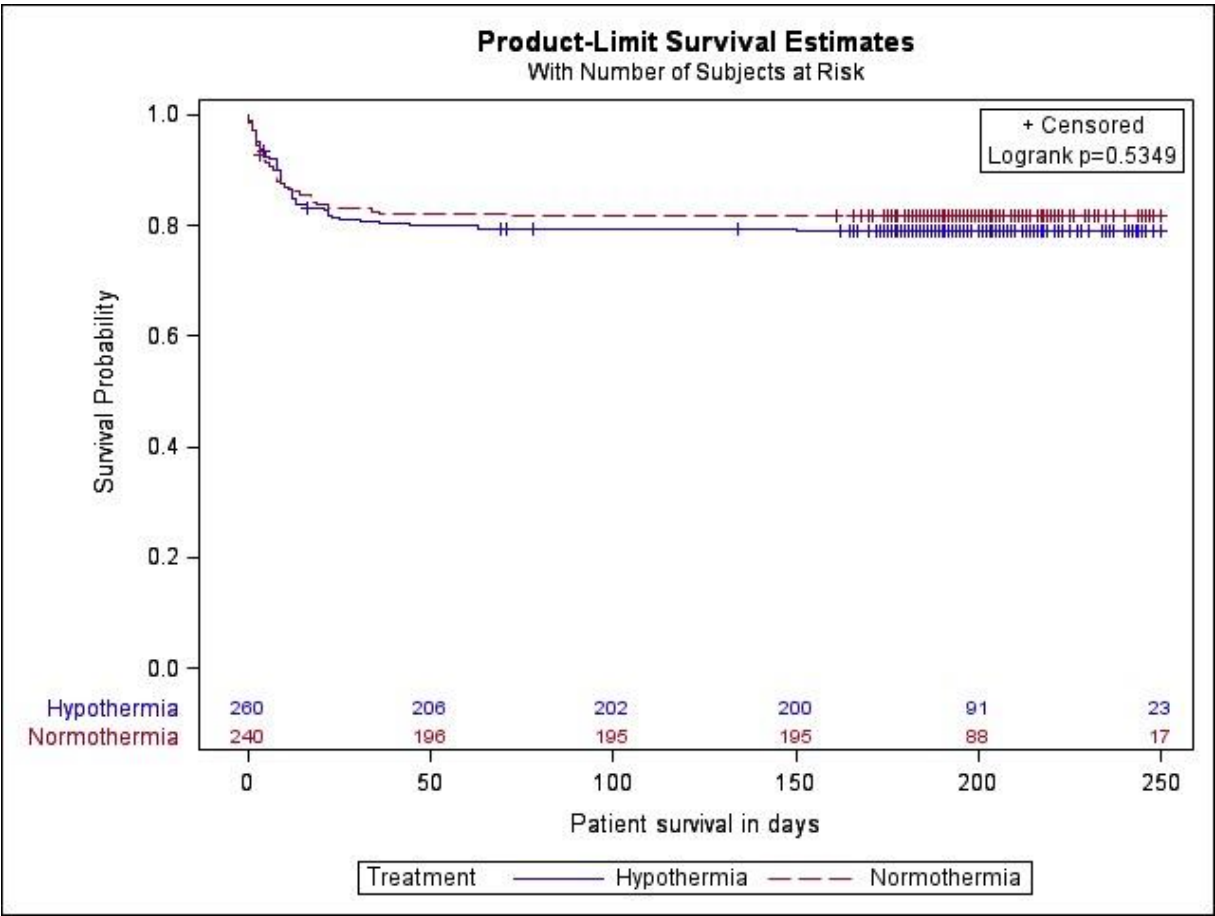
Abbreviations: GCS, Glasgow Coma Scale; TBI, traumatic brain injury.

eFigure 1. Hourly Temperature and ICP During Rewarming (36 hours) in the Hypothermia Group



Shown are least square means and 95% confidence intervals derived from longitudinal mixed linear modelling with each hypothermia patient treated as a random effect.

eFigure 2. Kaplan-Meier Survival Curve



Kaplan–Meier estimates of the unadjusted probability of death at 6 months in patients receiving normothermia or hypothermia.
Tick marks represent censored patients. Comparison of risk of death at 6 months (hypothermia vs normothermia) is given by an unadjusted hazard ratio 1.13 (0.76-1.69); $P = .54$.

eTable 2. Randomization Location and Times

	No. (%)		<i>P</i> Value
	Hypothermia (n = 260)	Normothermia (n = 240)	
Place of Randomization			.24
Out-of-hospital	123 (47.3)	101 (42.1)	
Emergency Department	137 (52.7)	139 (57.9)	
Times			
Injury to randomization, out-of-hospital, median (IQR), h	1.0 (0.8-1.3)	1.0 (0.7-1.4)	.98
Injury to randomization, hospital, median (IQR), h	2.6 (2.0-3.0)	2.6 (2.2-3.0)	.58
Patients who reached 35°C	233 (89.6)		
Injury to hypothermia (35°C), median (IQR), h	2.5 (0.8-5.5)	NA	
Patients who reached 33°C	186 (71.5)		
Injury to hypothermia (33°C), median (IQR), h	10.1 (6.8-15.9)	NA	
Abbreviations: IQR, interquartile range; NA, not applicable.			

eTable 3. Protocol Violations

	No. (%)	
	Hypothermia (n = 260)	Normothermia (n = 240)
Assigned to Hypothermia		
Withdrawn from hypothermia therapy	99 (38.1)	
Bleeding ^a	9 (3.5)	
TBI assessed as not severe (e.g. alcohol affected)	51 (19.6)	
Physician decision	35 (13.5)	
Serious adverse event	1 (0.4)	
Other	3 (1.2)	
Assigned to Normothermia		
Received non-trial mild hypothermia ^b		23 (9.6)

Abbreviation: TBI, traumatic brain injury.

We report withdrawal from hypothermia therapy at any time in the first 96 hours following randomization.

^a Clinically significant bleeding and hemostatic intervention.

^b Below 35°C for >1 hour.

eTable 4. Processes of Care

	No. (%)		P Value
	Hypothermia (n = 260)	Normothermia (n = 240)	
Intracranial pressure monitor (ICP)			
ICP monitor inserted	205 (78.8)	192 (80.0)	.75
Duration of monitoring, median (IQR), d	7.2 (5.0 to 9.6)	6.3 (3.6 to 9.4)	.04
End hours of ICP >20 mm Hg, median (IQR), h	6 (1 to 17)	6 (1 to 24)	.46
ICP index, median (IQR), % ^a	4.2 (1.0 to 11.5)	4.5 (1.3 to 13.7)	.60
End hours CPP ≤60 mm Hg, median (IQR), h	10 (3 to 18)	10 (3 to 22)	.69
Cerebral perfusion pressure (CPP) index, median (IQR), % ^b	6.5 (2.7 to 13.5)	7.4 (2.3 to 15.5)	.52
Therapies for increased ICP (Total for ICU stay)			
Hypertonic saline received	133 (51.2)	131 (54.6)	.44
Hypertonic saline (standardized to 3%), median (IQR), ml	1092 (359 to 2347)	1248 (546 to 2830)	.18
Mannitol 20% received	68 (26.2)	70 (29.2)	.45
Mannitol 20%, median (IQR), ml	375 (250 to 500)	250 (200 to 500)	.20
Thiopentone received	53 (20.4)	59 (24.6)	.26
Thiopentone total, median (IQR), g	7 (1.3 to 13.3)	3.3 (0.5 to 12.4)	.26
Therapies for increased ICP (First 10 ICU days)			
Cerebrospinal fluid venting	72 (27.7)	67 (27.9)	.96
PaCO ₂ <33 mm Hg	202 (77.7)	184 (76.7)	.79
PaCO ₂ <33 mm Hg, median (IQR), d	4 (2 to 5)	3 (2 to 5)	.64
Noradrenaline >15 mcg/min	145 (55.8)	143 (59.6)	.39
Noradrenaline >15 mcg/min, median (IQR), d	4 (2 to 6)	2 (2 to 4)	<.000
Adrenaline >5 mcg/min	17 (6.5)	9 (3.8)	.16
Adrenaline >5 mcg/min, median (IQR), d	1 (1 to 2)	1 (1 to 2)	.90
Neurosurgical procedures			
Surgical evacuation of subdural hematoma	56 (21.5)	51 (21.3)	.94
Surgical evacuation of extradural hematoma	27 (10.4)	31 (12.9)	.38
Surgical evacuation of intracerebral hemorrhage	4 (1.5)	6 (2.5)	.44
Craniectomy	56 (21.5)	60 (25.0)	.36
Decompressive craniectomy bifrontal	8 (3.1)	7 (2.9)	.92
Fluid management (First 10 ICU days)			
Albumin received (4-5%)	21 (8.1)	22 (9.2)	.66
Albumin (4-5%), median (IQR), ml ^c	1000 (500 to 1500)	500 (500 to 1000)	.54
Red-blood cells received	82 (31.5)	82 (34.2)	.53
Red-blood cells, median (IQR), ml ^c	753 (512 to 1342)	900 (565 to 1500)	.21
Fresh frozen plasma received	58 (22.3)	62 (25.8)	.36
Fresh frozen plasma, median (IQR), ml ^c	1058 (600 to 1791)	973 (600 to 1673)	.75
Fluid balance day 1, median (IQR), ml	2139 (1000 to 3501)	1897 (635 to 3669)	.29
Fluid balance days 1-5, median (IQR), ml	2100 (994 to 3441)	1900 (640 to 3687)	.44
Fluid balance days 6-10, median (IQR), ml	-505 (-1319 to 301)	-240 (-1116 to 616)	.04

^a ICP index: number of hours ICP documented as >20 mm Hg/number of hours ICP monitored

^b CPP index: number of hours CPP documented ≤60 mm Hg/number of hours CPP monitored

^c Total ml for first 10 days

eTable 5. Other Outcomes

	Hypothermia (n = 260)	Normothermia (n = 240)	P Value
Mechanical ventilation, median (IQR), d	8.3 (3.6-12.5)	6.9 (2.8-11.8)	.06
ICU length of stay, median (IQR), d	11.9 (5.7-18.3)	11.4 (4.7-17.5)	.33
Hospital length of stay, median (IQR), d	19.8 (10.2-31.8)	18.8 (8.1-32.8)	.74
GOS-E score at 6 months ^a			
Score, mean (SD)	4.2 (2.2)	4.2 (2.2)	.86
Unfavorable score of 2-4 in survivors, No. (%)	69 (37.1)	71 (39.0)	.71

Abbreviations: GOS-E, Glasgow Outcome Scale–Extended; IQR, interquartile range.

^a GOS-E score at 6 months was available for 240 patients in the hypothermia group and 226 in the normothermia group.

eTable 6. Adverse Events

	No. (%)		
	Hypothermia (n = 260)	Normothermia (n = 240)	P Value
Cardiovascular events			
Bradycardia (<40 beats per min)	49 (18.8)	10 (4.2)	<.0001
Ventricular tachycardia	7 (2.7)	5 (2.1)	.66
Ventricular fibrillation	1 (0.4)	0 (0)	1.00
Asystole/Pulseless electrical activity	7 (2.7)	7 (2.9)	.88
Hypotension on induction of hypothermia ^a	18 (6.9)	NA	
Hypotension during maintenance ^a	23 (8.8)	NA	
Infection			
Pneumonia	143 (55.0)	123 (51.3)	.40
Bacteremia	19 (7.3)	12 (5.0)	.29
Other infection (cerebral abscess, CNS, wound, other)	36 (13.8)	38 (15.8)	.53
Cerebral bleeding			
New or increased intracranial bleeding	47 (18.1)	37 (15.4)	.43
Non-cerebral bleeding			
New significant extra-cranial bleeding	8 (3.1)	6 (2.5)	.70
ICP, No./total No. (%)			
Intractable ICP ^b	43/260 (16.5)	43/239 (18.0)	.67

Abbreviations: CNS, central nervous system; ICP, intracranial pressure; NA, not applicable.

^a Episodes of mean arterial pressure <60 mm Hg.

^b Intractable ICP defined as ICP >20 mm Hg sustained continuously at this level for >20 minutes. Patient on maximal therapy or required substantive increase in therapy for ICP management.

eTable 7. Time to Reach 33°C for Hypothermia Patients (Approximate Tertiles)

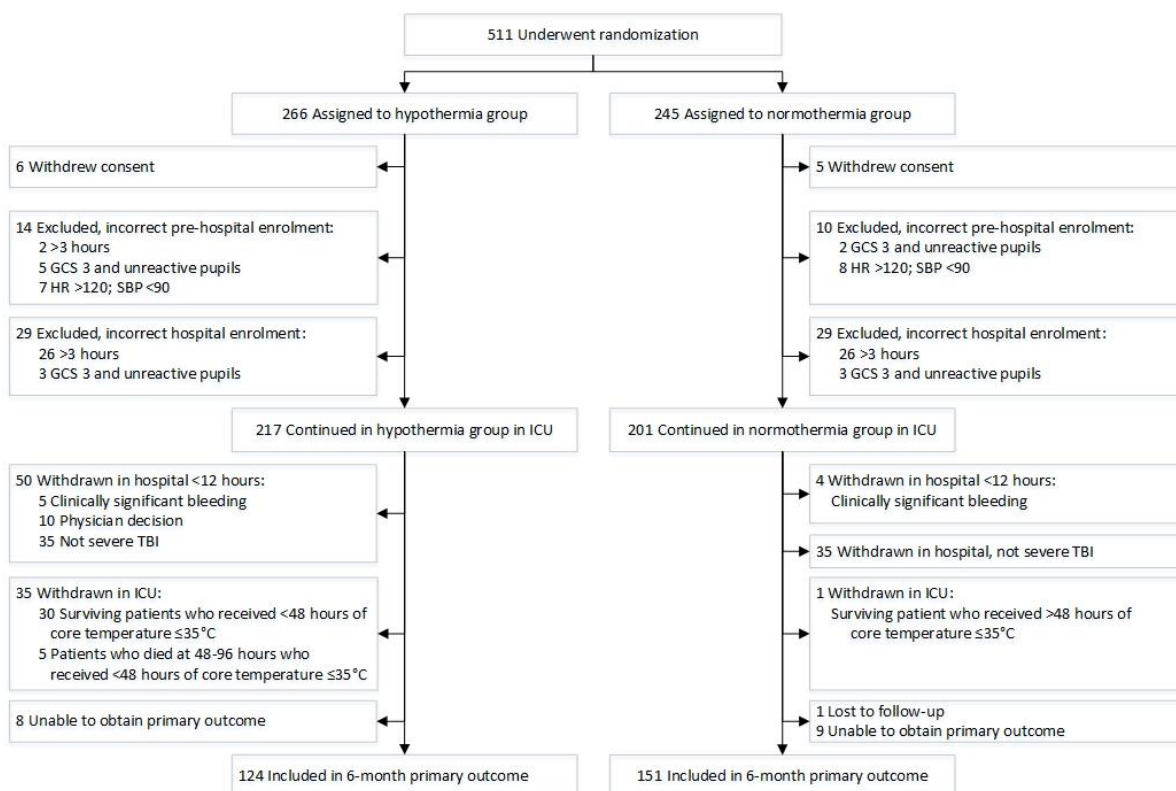
Time to 33°C	n	GOS-E Favorable Score No. [% (95% CI)]	P Value
1-7, h	49	18 [36.7 (22.8-50.7)]	.09
8-13, h	67	28 [41.8 (29.6-53.9)]	
>13, h	59	32 [54.2 (41.2-67.3)]	
Never	65	39 [60.0 (47.8-72.2)]	

Abbreviation: GOS-E, Glasgow Outcome Scale–Extended.

2. PER-PROTOCOL ANALYSIS

The per-protocol analysis was conducted as detailed in the SAP update.¹ The per-protocol analysis included hypothermia and normothermia patients who were demonstrated retrospectively to have suffered a severe TBI, thus excluding those with initially low GCS <9 not due to severe head injury but rather associated with transient influences, such as alcohol or drug effects. In addition, all patients who did not meet an inclusion criteria or who had an exclusion criteria present were excluded from the per-protocol analysis. Compliance with cooling in the per-protocol subset was defined as patients receiving >48 hours of cooling to $\leq 35^{\circ}\text{C}$ within 96 hours of randomization. Patients who survived at least 96 hours but who did not meet the compliance with cooling definition were excluded from the per-protocol analysis subset. Patients who did not survive 96 hours but who were known to be noncompliant at the time of their death were excluded. Patients who did not survive 96 hours but who were known to be compliant at their time of death, or whose compliance status was not completely determined, were included in the per protocol analysis subset.

eFigure 3. Patients Included in Per-Protocol Analysis



eTable 8. Per-Protocol Baseline

	No. (%)		P Value
	Hypothermia (n = 132)	Normothermia (n = 161)	
Men	107 (81.1)	132 (82.0)	.84
Women	25 (18.9)	29 (18.0)	
Age, mean (SD), y	35.2 (12.9)	34.4 (13.0)	.57
GCS overall score, median (IQR) ^a	5 (4-7)	6 (4-7)	.35
GCS motor score, median (IQR)	3 (1-4)	3 (2-5)	.25
One or both pupils reacting ^b	113 (85.6)	137 (85.1)	.90
Hypotension (out-of-hospital or ED), No./total No.	10/130 (7.7)	12/160 (7.5)	.95
Hypoxia (out-of-hospital or ED), No./total No. (%) ^d	18/129 (14.0)	25/159 (15.7)	.68
Temperature at the scene, mean (SD), °C ^e	35.9 (1.1)	35.9 (1.0)	.98
CT Marshall classification ^f			.73
Diffuse injury I (normal findings)	2 (1.5)	2 (1.2)	
Diffuse injury II	72 (54.5)	84 (52.2)	
Diffuse injury III or IV	14 (10.6)	15 (9.3)	
Evacuated mass lesion V	44 (33.3)	58 (36.0)	
Non-evacuated mass lesion VI	0 (0)	2 (1.2)	
Probability of unfavorable outcome at 6 months: IMPACT-TBI (core + CT), mean (SD) ^g	0.48 (0.23)	0.47 (0.21)	.74
Injury Severity Score, median (IQR) ^h	29 (21-34.5)	29 (22-35)	.44
Cause of injury			.45
Motor vehicle	43 (32.6)	59 (36.6)	
Motorcycle	14 (10.6)	12 (7.5)	
Bicycle	12 (9.1)	14 (8.7)	
Pedestrian	15 (11.4)	28 (17.4)	
Hit by object	13 (9.8)	10 (6.2)	
Fall/jump	28 (21.2)	34 (21.1)	
Other	7 (5.3)	4 (2.5)	
Positive blood ethanol level, No./total No. (%)	51/111 (45.9)	58/130 (44.6)	.84
Blood ethanol >51 mg/dL, No./total No. (%)	38/111 (34.2)	43/130 (33.1)	.85
Time from injury to randomization, median (IQR), h	1.7 (1.0-2.5)	2.0 (1.1-2.6)	.18

Abbreviations: CT, computed tomography; ED, emergency department; GCS, Glasgow Coma Scale; IMPACT-TBI, International Mission for Prognosis and Analysis of Clinical Trials in Traumatic Brain Injury; IQR, interquartile range.

^a The highest reliable score before randomization is reported; overall scores on the GCS range from 3 to 15, with lower scores indicating a lower level of consciousness. A patient with a GCS score of 6 is unconscious and will likely have a flexion response to pain.

^b Some patients had a GCS score greater than 3, with small unreactive pupils.

^c Hypotension was defined as a systolic blood pressure less than 90 mm Hg.

^d Hypoxia was defined as SpO₂ less than 90%.

^e Temperature at the scene was available for 78 patients in the hypothermia group and 83 in the normothermia group.

^f The Marshall classification of CT abnormalities in brain trauma ranges from I to VI, with a score of I indicating normal findings, II diffuse injury, III or IV radiologic signs of increased intracranial pressure, and V or VI an intracranial mass lesion. The first CT scan was categorized for each patient. Patients who had surgery to evacuate a hematoma before 24 hours after injury but whose first CT was conducted before surgery were classified as having Marshall score V.

^g IMPACT-TBI score is validated to predict the outcome of patients with a head injury and a GCS score less than 13. It provides the predicted probability of a 6-month poor outcome (Glasgow Outcome Scale–Extended score of ≤4) ranging from 0.0 to 1.0, in which 1.0 represents 100% and considers age, motor score, pupil response, hypoxia, hypotension, and CT classification.

^h The Injury Severity Score ranges from 0 to 75, with higher scores indicating greater severity of injury. A score of 27 indicates severe injury and multitrauma.

eTable 9. Per-Protocol Analysis^a

	No./Total No. (%)			
Outcome	Hypothermia	Normothermia	Relative Risk (95% CI)	P Value
Primary Outcome				
Favorable outcome at 6 months (GOS-E score 5-8)	58/124 (46.8)	69/151 (45.7)	1.02 (0.79-1.32)	.86
Secondary Outcomes				
Severity-adjusted relative risk for favorable outcome (using IMPACT-TBI)			1.04 (0.65-1.69)	.86
Death in the hospital	30/132 (22.7)	33/160 (20.6)	1.10 (0.71-1.71)	.66
Death at 6 months	31/132 (23.5)	34/160 (21.3)	1.11 (0.72-1.70)	.65
Infections				
Pneumonia	93/132 (70.5)	92/161 (57.1)	1.23 (1.04-1.47)	.02
Bacteremia	14/132 (10.6)	8/161 (5.0)	2.13 (0.92-4.93)	.07
Other infection	25/132 (18.9)	30/161 (18.6)	1.02 (0.63-1.64)	.95
Bleeding				
New or increased intracranial bleeding	26/132 (19.7)	29/161 (18.0)	1.09 (0.68-1.76)	.71
New significant extra-cranial bleeding	2/132 (1.5)	6/161 (3.7)	0.41 (0.08-1.98)	.25
Subgroups, Favorable Outcome (GOS-E Score 5-8)				
Surgically removed hematomas ^b				
Yes	12/41 (29.3)	20/54 (37.0)	0.79 (0.44-1.42)	.43
No	46/83 (55.4)	49/97 (50.5)	1.10 (0.83-1.45)	.51
Any intracranial mass lesion (Marshall classification V + VI) ^c				
Yes	12/41 (29.3)	21/56 (37.5)	0.78 (0.44-1.40)	.40
No	46/83 (55.4)	48/95 (50.5)	1.10 (0.83-1.45)	.51

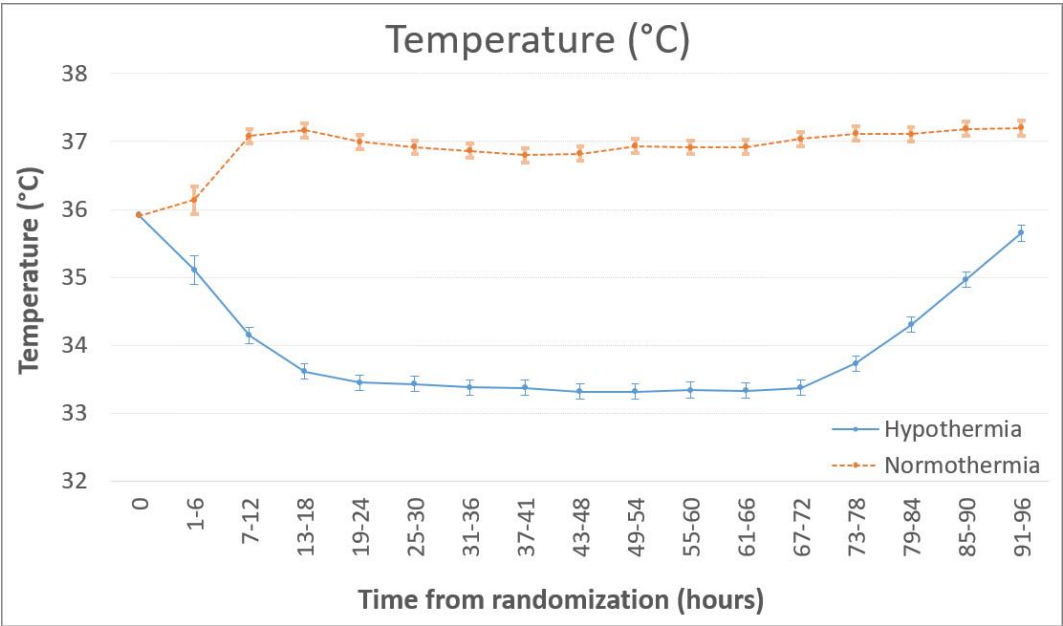
Abbreviations: GOS-E, Glasgow Outcome Scale–Extended; IMPACT-TBI, International Mission for Prognosis and Analysis of Clinical Trials in Traumatic Brain Injury.

^a The per-protocol analysis was conducted as detailed in the Statistical Analysis Plan Update.²

^b P value for interaction = .31.

^c P value for interaction = .29.

eFigure 4. Hourly Temperature for First 4 days (96 hours) Post-Randomization in the Per-Protocol Analysis of Patients

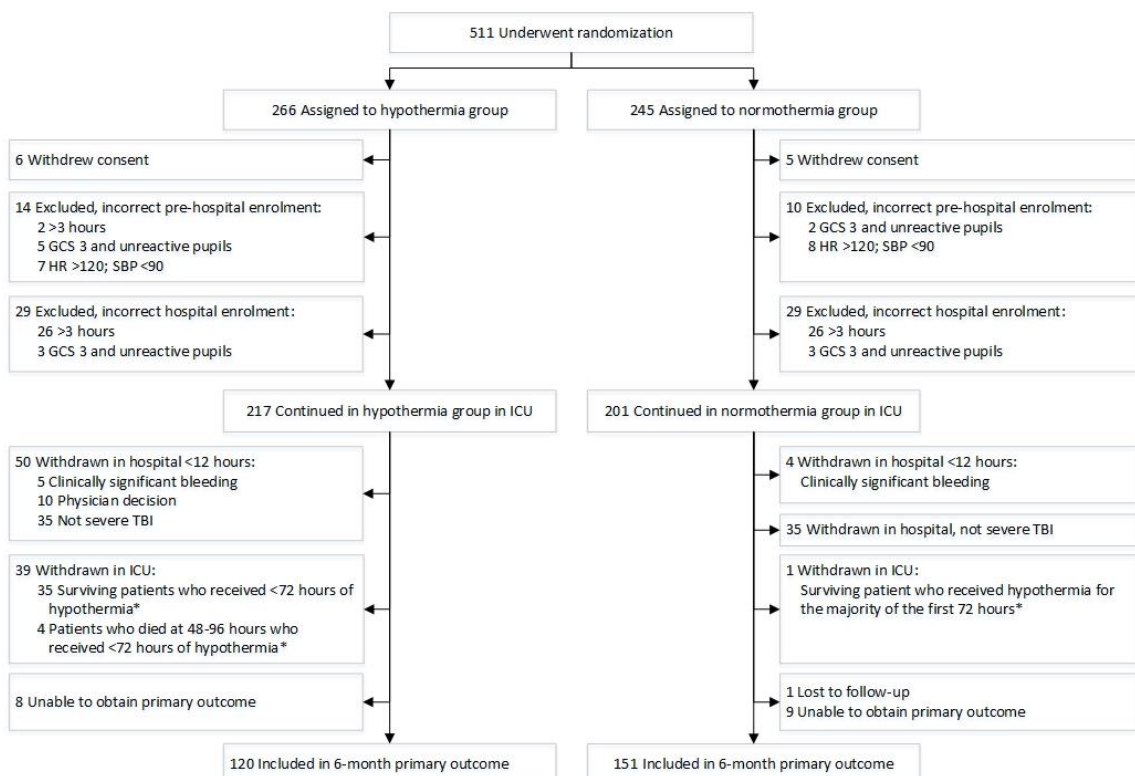


PER-PROTOCOL SENSITIVITY ANALYSIS

To increase robustness of the interpretation of patients that were deemed to be compliant with cooling, an additional sensitivity analysis was performed with compliance defined as patients that were cooled for the majority of their first 72 hours instead of 96 hours. This was calculated using the area (degree × hours) under the protocol-defined cooling baseline value of 35 °C within 72 hours of randomisation. Each individual's AUC was estimated by the linear trapezoidal method applied to temperature data ignoring missing values, as detailed in the SAP.²

The agreement between the per-protocol and per-protocol sensitivity analysis methods was high [Kappa (95% CI) 0.93 (0.90-0.97)].

eFigure 5. Patients Included in Per-Protocol – Sensitivity Analysis



* Patients who were considered non-compliant with the cooling protocol were excluded. Compliance was determined based on whether a patient received hypothermia for the majority of their first 72 hours (defined using the area under the protocol-defined cooling baseline value of 35°C within 72 hours of randomization).^{1,2}

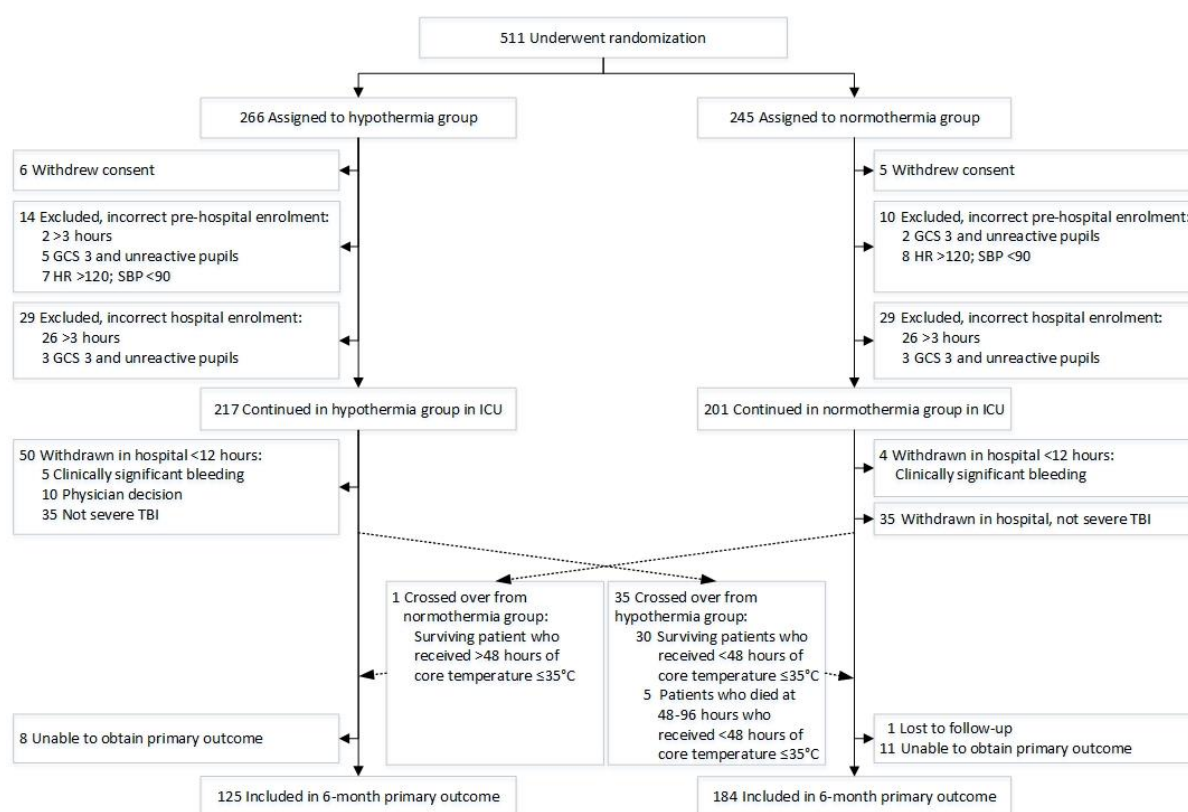
PER-PROTOCOL SENSITIVITY ANALYSIS – PRIMARY OUTCOME

Favorable outcomes (GOS-E score 5-8) at 6 months occurred in 55/120 patients (45.8%) in the hypothermia group and in 69/151 patients (45.7%) in the normothermia group (unadjusted relative risk with hypothermia of 1.00 [95% CI, 0.77 to 1.30], $P = .98$).

3. AS-TREATED ANALYSIS

The as-treated analysis was conducted as detailed in the SAP update.¹ The as-treated analysis included hypothermia and normothermia patients who were demonstrated retrospectively to have suffered a severe TBI, thus excluding those with initially low GCS score <9 not due to severe head injury but rather associated with transient influences, such as alcohol or drug effects. In addition, all patients who did not meet an inclusion criteria or who had an exclusion criteria present were excluded from the as-treated analysis subset. Compliance with cooling in the as-treated analysis subsets was defined as patients receiving >48 hours of cooling to $\leq 35^{\circ}\text{C}$ within 96 hours of randomization. Patients who met the compliance with cooling criteria (regardless of randomization category) were analysed in the hypothermia group. Patients who did not meet the definition of the compliance with cooling were analysed with the normothermia group. Patients who do not survive 96 hours but who were known to be noncompliant with their assigned treatment at the time of their death were analysed in the opposite treatment arm to their assignment. Patients who did not survive 96 hours but who were known to be compliant at their time of death, or whose compliance status was not completely determined, were analysed in their assigned treatment arm.

eFigure 6. Patients Included in As-Treated Analysis



eTable 10. As-Treated Baseline

	No. (%)		P Value
	Hypothermia (n = 133)	Normothermia (n = 196)	
Men	108 (81.2)	158 (80.6)	.89
Women	25 (18.8)	38 (19.4)	
Age, mean (SD), y	35.2 (12.8)	34.3 (13.2)	.55
GCS overall score, median (IQR) ^a	5 (4-7)	6 (4-7)	.29
GCS motor score, median (IQR)	3 (1-4)	4 (2-5)	.23
One or both pupils reacting ^b	114 (85.7)	167 (85.2)	.90
Hypotension (out-of-hospital or ED), No./total No. (%) ^c	10/131 (7.6)	15/194 (7.7)	.97
Hypoxia (out-of-hospital or ED), No./total No. (%) ^d	18/130 (13.8)	30/194 (15.5)	.69
Temperature at the scene, mean (SD), °C ^e	35.9 (1.1)	35.9 (1.0)	.87
CT Marshall classification ^f			.71
Diffuse injury I (normal findings)	2 (1.5)	3 (1.5)	
Diffuse injury II	73 (54.9)	108 (55.1)	
Diffuse injury III or IV	14 (10.5)	15 (7.7)	
Evacuated mass lesion V	44 (33.1)	34.7 (68)	
Non-evacuated mass lesion VI	0 (0)	1 (2)	
Probability of unfavorable outcome at 6 months: IMPACT-TBI (core + CT), mean (SD) ^g	0.48 (0.23)	0.46 (0.22)	.50
Injury Severity Score, median (IQR) ^h	29 (22-35)	29 (21-34)	.59
Cause of injury			.64
Motor vehicle	43 (32.3)	70 (35.7)	
Motorcycle	14 (10.5)	18 (9.2)	
Bicycle	12 (9)	16 (8.2)	
Pedestrian	15 (11.3)	31 (15.8)	
Hit by object	13 (9.8)	13 (6.6)	
Fall/jump	29 (21.8)	43 (21.9)	
Other	7 (5.3)	5 (2.6)	
Positive blood ethanol level, No./total No. (%)	52/112 (46.4)	69/152 (45.4)	.87
Blood ethanol >51 mg/dL, No./total No. (%)	39/112 (34.8)	52/152 (34.2)	.92
Time from injury to randomization, median (IQR), h	1.7 (1.0-2.5)	1.9 (1.0-2.6)	.37

Abbreviations: CT, computed tomography; ED, emergency department; GCS, Glasgow Coma Scale; IMPACT-TBI, International Mission for Prognosis and Analysis of Clinical Trials in Traumatic Brain Injury; IQR, interquartile range.

^a The highest reliable score before randomization is reported; overall scores on the GCS range from 3 to 15, with lower scores indicating a lower level of consciousness. A patient with a GCS score of 6 is unconscious and will likely have a flexion response to pain.

^b Some patients had a GCS score greater than 3, with small unreactive pupils.

^c Hypotension was defined as a systolic blood pressure less than 90 mm Hg.

^d Hypoxia was defined as SpO₂ less than 90%.

^e Temperature at the scene was available for 78 patients in the hypothermia group and 99 in the normothermia group.

^f The Marshall classification of CT abnormalities in brain trauma ranges from I to VI, with a score of I indicating normal findings, II diffuse injury, III or IV radiologic signs of increased intracranial pressure, and V or VI an intracranial mass lesion. The first CT scan was categorized for each patient. Patients who had surgery to evacuate a hematoma before 24 hours after injury but whose first CT was conducted before surgery were classified as having Marshall score V.

^g IMPACT-TBI score is validated to predict the outcome of patients with a head injury and a GCS score less than 13. It provides the predicted probability of a 6-month poor outcome (Glasgow Outcome Scale–Extended score of ≤4) ranging from 0.0 to 1.0, in which 1.0 represents 100% and considers age, motor score, pupil response, hypoxia, hypotension and CT classification.

^h The Injury Severity Score ranges from 0 to 75, with higher scores indicating greater severity of injury. A score of 27 indicates severe injury and multitrauma.

eTable 11. As-Treated Analysis^a

	<u>No./Total No. (%)</u>		Relative Risk (95% CI)	P Value
Outcome	Hypothermia	Normothermia		
Primary Outcome				
Favorable outcome at 6 months (GOS-E score 5-8)	59/125 (47.2)	82/184 (44.6)	1.06 (0.83-1.35)	.65
Secondary Outcomes				
Severity-adjusted relative risk for unfavorable outcome (using IMPACT-TBI)			1.11 (0.70-1.76)	.65
Death in the hospital	30/133 (22.6)	38/195 (19.5)	1.16 (0.76-1.77)	.50
Death at 6 months	31/133 (23.3)	40/195 (20.5)	1.14 (0.75-1.72)	.55
Infections				
Pneumonia	94/133 (70.7)	107/196 (54.6)	1.29 (1.09-1.53)	.003
Bacteremia	14/133 (10.5)	11/196 (5.6)	1.88 (0.88-4.00)	.10
Other infection	26/133 (19.5)	33/196 (16.8)	1.16 (0.73-1.85)	.53
Bleeding				
New or increased intracranial bleeding	26/133 (19.5)	39/196 (19.9)	0.98 (0.63-1.53)	.94
New significant extra-cranial bleeding	2/133 (1.5)	10/196 (5.1)	0.29 (0.07-1.32)	.09
Subgroups, Favorable Outcome (GOS-E Score 5-8)				
Surgically removed hematomas ^b				
Yes	12/41 (29.3)	23/64 (35.9)	0.81 (0.46-1.45)	.48
No	47/84 (56.0)	59/120 (49.2)	1.14 (0.87-1.48)	.34
Any intracranial mass lesion (Marshall classification V + VI) ^c				
Yes	12/41 (29.3)	24/66 (36.4)	0.80 (0.45-1.43)	.45
No	47/84 (56.0)	58/118 (49.2)	1.14 (0.87-1.48)	.34

Abbreviations: GOS-E, Glasgow Outcome Scale–Extended score; IMPACT-TBI, International Mission for Prognosis and Analysis of Clinical Trials in Traumatic Brain Injury.

^a The as-treated analysis was conducted as detailed in the Statistical Analysis Plan Update.¹

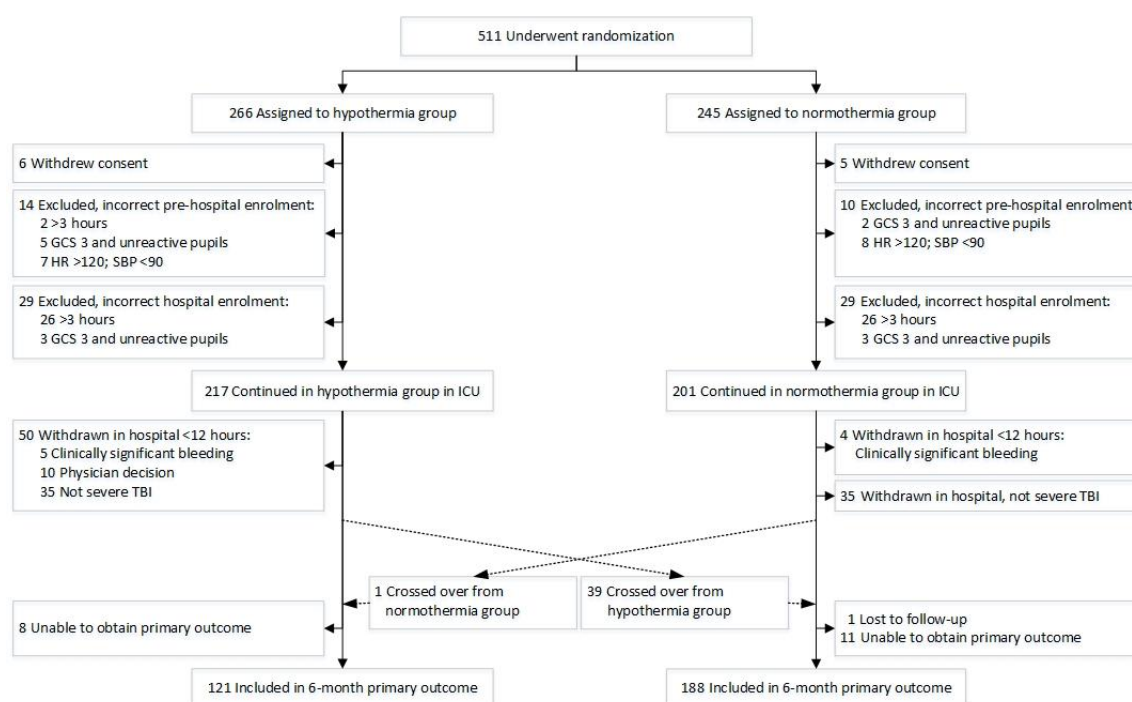
^b P value for interaction = .27.

^c P value for interaction = .25.

AS-TREATED SENSITIVITY ANALYSIS

To increase robustness of the interpretation of patients that were deemed to be compliant with cooling, an additional sensitivity analysis was performed with compliance defined as patients that were cooled for the majority of their first 72 hours instead of 96 hours. This was calculated using the area (degree × hours) under the protocol-defined cooling baseline value of 35 °C within 72 hours of randomisation. Each individual's AUC was estimated by the linear trapezoidal method applied to temperature data ignoring missing values, as detailed in the SAP.² Patients in the hypothermia group who did not receive hypothermia for the majority of their first 72 hours were analyzed with the normothermia group. Patients in the normothermia group who received hypothermia for the majority of their first 72 hours for rescue therapy were analyzed with the hypothermia group.

eFigure 7. Patients Included in As-Treated – Sensitivity Analysis



AS-TREATED SENSITIVITY ANALYSIS – PRIMARY OUTCOME

Favorable outcomes (GOS-E score 5-8) at 6 months occurred in 56/121 patients (46.3%) in the hypothermia group and in 85/188 patients (45.2%) in the normothermia group (unadjusted relative risk with hypothermia of 1.02 [95% CI, 0.80 to 1.31], $P = .85$).

4. POST-HOC ANALYSES

POST-HOC SENSITIVITY ANALYSIS OF MISSINGNESS

6.8% (34/500) of patients were missing the primary outcome (GOS-E score of 5 to 8), with 7.7% (20/260) missing from the hypothermia group and 5.8% (14/240) missing from the normothermia group. To evaluate the impact of missingness on the primary outcome, sensitivity analysis was conducted using the following 3 approaches:

1. Worst case scenario for hypothermia: all missing patients from the hypothermia group were assigned a bad outcome while all missing patients from the normothermia group were assigned a good outcome.
2. Best case scenario for hypothermia: all missing patients from the hypothermia group were assigned a good outcome (GOS-E score 5 to 8) while all missing patients from the normothermia group were assigned a bad outcome.
3. Most likely scenario for hypothermia using multiple imputation: missingness for the primary outcome was conditional on observed covariates and was assumed to be “missing at random”. Predictive mean matching was employed using all available baseline and hospital discharge data with a total of 11 imputed datasets generated.

Results from scenarios 1 and 2 have been compared using unadjusted chi-square test for equal proportions, and reported as frequency (percentage) per treatment group with relative risk and risk difference, both accompanied by 95% confidence intervals. Results from scenario 3 have been compared using a Cochran–Mantel–Haenszel test for stratified proportions and reported as a relative risk with risk difference computed from Miettinen-Nurminen (score) confidence limits for the stratum risk differences.

eTable 12. Sensitivity Analysis of Missingness

Outcome Scenario	No. [% (95% CI)]		(95% CI)		
	Hypothermia (n = 260)	Normothermia (n = 240)	Relative Risk	Risk Difference	P Value
1. Worst case	117 [45.0 (38.8 to 51.2)]	125 [52.1 (45.6 to 58.5)]	0.86 (0.72 to 1.04)	-7.1 (-15.8 to 1.7)	.11
2. Best case	137 [52.7 (46.5 to 58.9)]	111 [46.3 (39.8 to 52.7)]	1.14 (0.95 to 1.36)	6.4 (-2.3 to 15.2)	.15
3. Most likely	133 [51.1 (45.1 to 57.2)]	121 [50.4 (44.1 to 56.7)]	1.00 (0.95 to 1.06)	0.7 (-8.0 to 9.5)	.88

POST-HOC ANALYSES OF EVALUABLE HYPOTHERMIA PATIENTS THAT RECEIVED ADEQUATE COOLING VS NORMOTHERMIA

To ensure that the patients who completed the hypothermia protocol were still comparable to the patients randomized to the control group, an additional post-hoc analysis was performed of evaluable patients who received an adequate dose of hypothermia (>48hrs) compared to controls. Categorical variables were compared using chi-square tests for equal proportion, student t-tests for normally distributed data and Wilcoxon rank sum tests otherwise with results reported as n (%), mean (standard deviation) or median (interquartile range) respectively. Where potential imbalances were found between groups, hierarchical multivariable logistic regression was employed adjusting for baseline imbalance and treating randomization strata (location and site) as random effects with results reported as odds ratios (95% CI).

RESULTS

When comparing control patients to evaluable hypothermia patients who received adequate cooling, groups were similar (eTable 13), with the only significant baseline imbalance being time from injury to randomization with control patients randomised on average 16 minutes later than hypothermia patients [1.73 (0.78) hours vs 2.00 (1.04) hours, $P = .01$]. When considering the proportion of patients with a favorable outcome, there was no significant difference between groups [hypothermia 46.8% (58/124) vs normothermia 49.1% (111/226); odds ratio (95% CI) 0.91 (0.59-1.41) $P = .68$]. When adjusting favorable outcome for time to randomization, the underlying result did not change [adjusted odds ratio hypothermia vs normothermia; 0.95 (0.55-275 1.64) $P = .84$].

eTable 13. Comparison of Baseline Characteristics for Patients who Completed the Hypothermia Protocol and had Primary Outcome vs All Controls with Primary Outcome

	No. (%)		P Value
	Compliant Hypothermia (n = 124)	Normothermia (n = 226)	
Men	100 (80.6)	180 (79.6)	.82
Women	24 (19.4)	46 (20.4)	
Age, mean (SD), y	34.7 (12.7)	34.3 (13.6)	.78
GCS overall score, median (IQR) ^a	5 (4-7)	6 (4-7)	.39
GCS motor score, median (IQR)	3 (1-4)	3 (2-5)	.43
One or both pupils reacting ^b	106 (85.5)	188 (83.2)	.58
Hypotension (out-of-hospital or ED), No./total No. (%) ^c	10/123 (8.1)	27/225 (12)	.26
Hypoxia (out-of-hospital or ED), No./total No. (%) ^d	17/121 (14)	34/223 (15.2)	.77
Temperature at the scene, mean (SD), °C ^e	35.9 (1.1)	35.9 (1.0)	.86
CT Marshall classification ^f			.28
Diffuse injury I (normal findings)	2 (1.6)	14 (6.2)	
Diffuse injury II	67 (54)	121 (53.5)	
Diffuse injury III or IV	14 (11.3)	20 (8.8)	
Evacuated mass lesion V	41 (33.1)	68 (30.1)	
Non-evacuated mass lesion VI	0 (0)	3 (1.3)	
IMPACT-TBI unfavorable outcome, mean (SD) ^g	0.48 (0.23)	0.47 (0.22)	.54
Injury Severity Score, median (IQR) ^h	29 (21-34.5)	29 (21-35)	.95
Cause of injury			.51
Motor vehicle	41 (33.1)	82 (36.3)	
Motorcycle	14 (11.3)	17 (7.5)	
Bicycle	11 (8.9)	20 (8.8)	
Pedestrian	14 (11.3)	35 (15.5)	
Hit by object	12 (9.7)	16 (7.1)	
Fall/jump	25 (20.2)	50 (22.1)	
Other	7 (5.6)	6 (2.7)	
Positive blood ethanol level, No./total No. (%)	48/105 (45.7)	80/180 (44.4)	.84
Blood ethanol >51 mg/dL, No./total No. (%)	35/105 (33.3)	60/180 (33.3)	1.00
Time from injury to randomization, median (IQR), hr	1.6 (1.0-2.5)	2.1 (1.1-2.8)	.02

Abbreviations: CT, computed tomography; ED, Emergency Department; GCS, Glasgow Coma Scale; IMPACT-TBI, International Mission for Prognosis and Analysis of Clinical Trials in Traumatic Brain Injury.

^a The highest reliable score before randomization is reported; overall scores on the GCS range from 3 to 15, with lower scores indicating a lower level of consciousness. A patient with a GCS score of 6 is unconscious and will likely have a flexion response to pain.

^b Some patients had a GCS score greater than 3, with small unreactive pupils.

^c Hypotension was defined as a systolic blood pressure less than 90 mm Hg.

^d Hypoxia was defined as SpO₂ less than 90%.

^e Temperature at the scene was available for 75 patients in the hypothermia group and 136 in the normothermia group.

^f The Marshall classification of CT abnormalities in brain trauma ranges from I to VI, with a score of I indicating normal findings, II diffuse injury, III or IV radiologic signs of increased intracranial pressure, and V or VI an intracranial mass lesion. The first CT scan was categorized for each patient. Patients who had surgery to evacuate a hematoma before 24 hours after injury but whose first CT was conducted before surgery were classified as having Marshall score V.

^g IMPACT-TBI score is validated to predict the outcome of patients with a head injury and a GCS score less than 13. It provides the predicted probability of a 6-month poor outcome (Glasgow Outcome Scale–Extended score of ≤4) ranging from 0.0 to 1.0, in which 1.0 represents 100% and considers age, motor score, pupil response, hypoxia, hypotension and CT classification.

^h The Injury Severity Score ranges from 0 to 75, with higher scores indicating greater severity of injury. A score of 27 indicates severe injury and multitrauma.

5. eREFERENCES

1. Cooper DJ. POLAR SAP Update: https://www.monash.edu/_data/assets/pdf_file/0006/1399434/2018-June-12_POLAR-SAP-Update.pdf and POLAR SAP Update Correction: https://www.monash.edu/_data/assets/pdf_file/0011/1400051/2018-June-13_POLAR-SAP-Update-Correction.pdf. 2018.
2. Presneill J, Gantner D, Nichol A, et al. Statistical analysis plan for the POLAR-RCT: The Prophylactic hypOthermia trial to Lessen trAumatic bRain injury-Randomised Controlled Trial. *Trials*. 2018;19(1):259.